Engineering Economy

Chapter 5:

Evaluating a Single Project

The objective of Chapter 5 is to discuss and **critique contemporary methods for determining project profita**bility.

Chapter 5
5.1. Introduction

All engineering economy studies of capital projects should consider the return that a given project will or should produce.

Proposed capital projects can be evaluated in several ways.

- Present worth (PW)
- Future worth (FW)
- Annual worth (AW)
- Internal rate of return (IRR)
- External rate of return (ERR)
- Payback period (generally not appropriate as a primary decision rule)

Chapter 5
5.2. Determining the Minimum Attractive Rate of Return

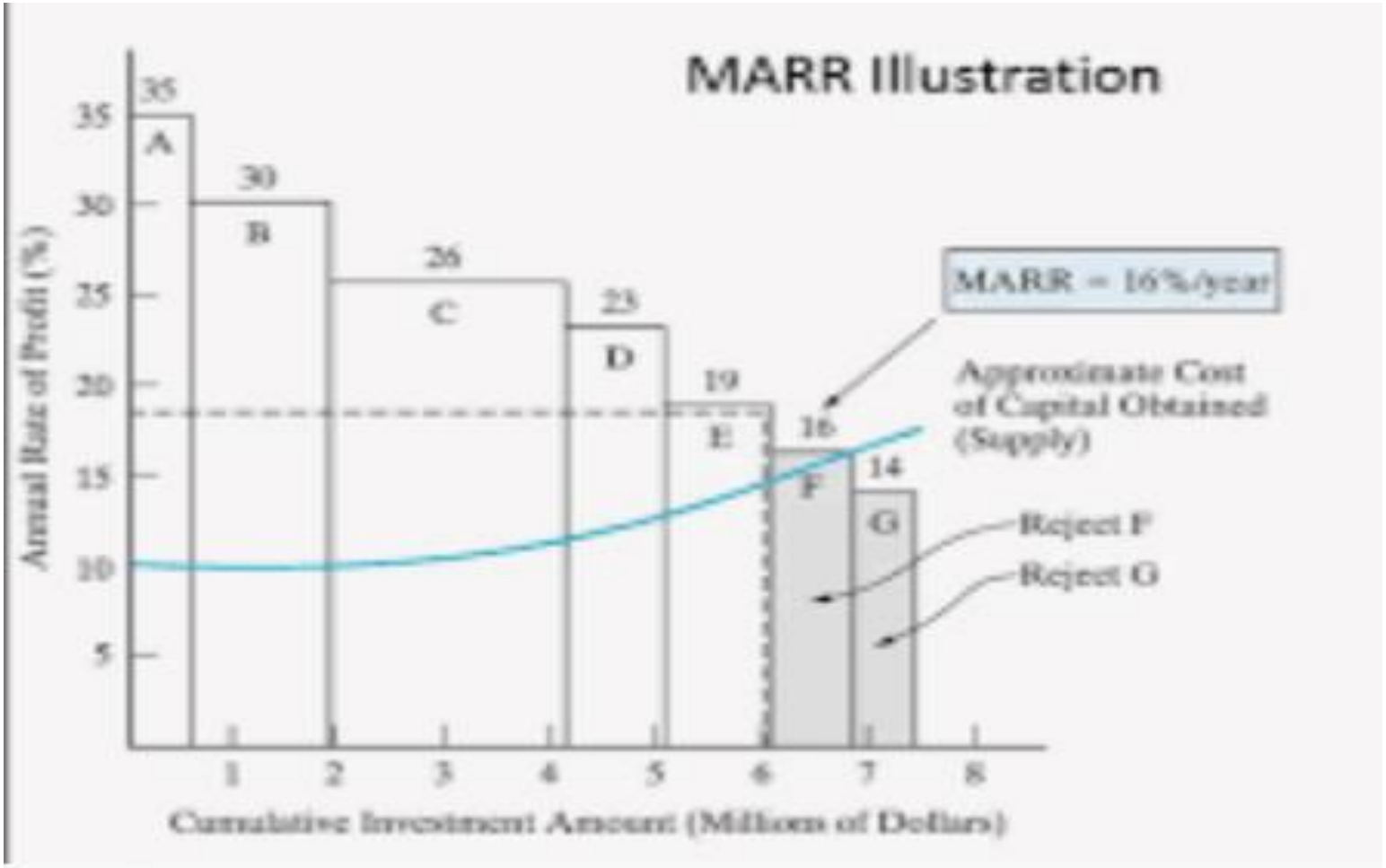
To be attractive, a capital project must provide a return that exceeds a minimum level established by the organization.

This minimum level is reflected in a firm's Minimum Attractive Rate of Return (MARR).

Many elements contribute to determining the MARR.

- Amount, source, and cost of money available
- Number and purpose of good projects available
- Perceived risk of investment opportunities
- Type of organization

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5.2. Determining the Minimum Attractive Rate of Return



Determinatin of Minimum Attractive Rate of Return (MARR).

Chapter 5
5.3. The Present Worth Method

The most-used method is the present worth method.

The present worth (PW) is found by discounting all cash inflows and outflows to the present time at an interest rate that is generally the MARR.

A positive PW for an investment project means that the project is acceptable (it satisfies the MARR).

To find the PW as a function of i% (per interest period) of a series of cash inflows and outflows, it is necessary to discount future amounts to the present by using the appropriate study period (years, for example) in the following manner:

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$$PW = F_0(1+i)^0 + F_1(1+i)^{-1} + F_2(1+i)^{-2} + \ldots + F_N(1+i)^{-N}$$
$$PW = \sum_{k=0}^{N} F_k(1+i)^{-k}$$

$i\%$	: effective interest rate, or MARR, per compounding period;
k	: index for each compounding period
F_k	: future cash flow at the end of period k;
N	: number of compounding periods in the planning horizon

To apply the PW method of determining a Project's economic worthiness, we simply compute the present equivalent of all cash flows using the MARR as the interest rate.
If the present worth is greater than or equal to zero, the Project is acceptable.

PW Decision Rule:

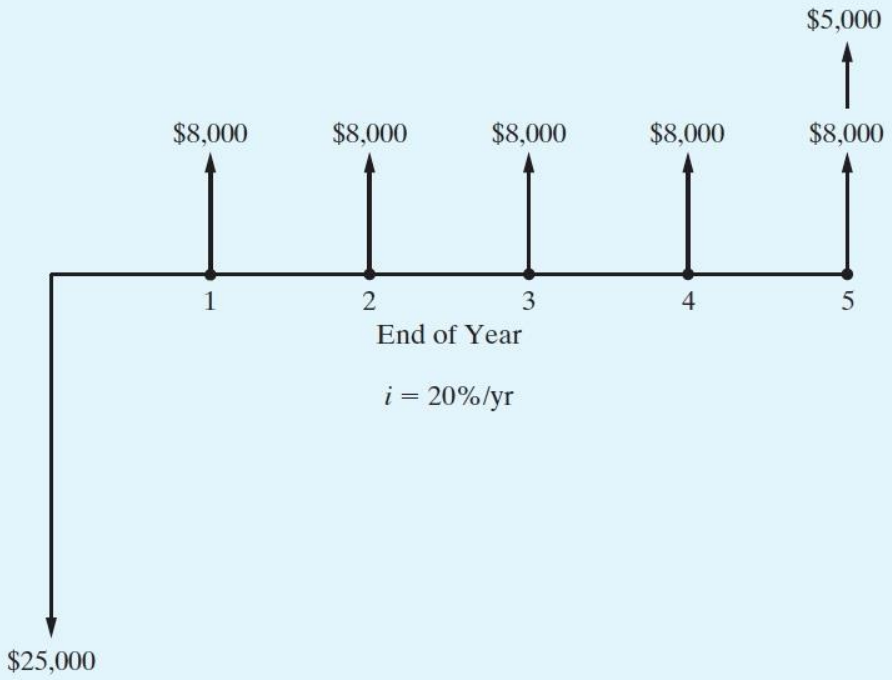
If PW (i=MARR) ≥ 0, the project is economically justified

Chapter 5
5.3. The Present Worth Method

EXAMPLE 5-1 **Evaluation of New Equipment Purchase Using PW**

VIDEO SOLUTIONS

A piece of new equipment has been proposed by engineers to increase the productivity of a certain manual welding operation. The investment cost is $25,000, and the equipment will have a market value of $5,000 at the end of a study period of five years. Increased productivity attributable to the equipment will amount to $8,000 per year after extra operating costs have been subtracted from the revenue generated by the additional production. A cash-flow diagram for this investment opportunity is given below. If the firm's MARR is 20% per year, is this proposal a sound one? Use the PW method.



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5.3. The Present Worth Method

EXAMPLE 5-1 **Evaluation of New Equipment Purchase Using PW**

Solution

$$\text{PW} = \text{PW of cash inflows} - \text{PW of cash outflows},$$

or

$$\begin{aligned}\text{PW}(20\%) &= \$8{,}000(P/A, 20\%, 5) + \$5{,}000(P/F, 20\%, 5) - \$25{,}000 \\ &= \$934.29.\end{aligned}$$

Because $\text{PW}(20\%) \geq 0$, this equipment is economically justified.

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5.3. The Present Worth Method

EXAMPLE 5-2 **Present Worth of a Space-Heating System**



A retrofitted space-heating system is being considered for a small office building. The system can be purchased and installed for $110,000, and it will save an estimated 300,000 kilowatt-hours (kWh) of electric power each year over a six-year period. A kilowatt-hour of electricity costs $0.10, and the company uses a MARR of 15% per year in its economic evaluations of refurbished systems. The market value of the system will be $8,000 at the end of six years, and additional annual operating and maintenance expenses are negligible. Use the PW method to determine whether this system should be installed.

Solution

To find the PW of the proposed heating system, we need to find the present equivalent of all associated cash flows. The estimated annual savings in electrical power is worth 300,000 kWh × $0.10/kWh = $30,000 per year. At a MARR of 15%, we get

$$\begin{aligned} \text{PW}(15\%) &= -\$110{,}000 + \$30{,}000\ (P/A, 15\%, 6) + \$8{,}000\ (P/F, 15\%, 6) \\ &= -\$110{,}000 + \$30{,}000(3.7845) + \$8{,}000(0.4323) \\ &= \$6{,}993.40. \end{aligned}$$

Since PW(15%) ≥ 0, we conclude that the retrofitted space-heating system should be installed.

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Example: Consider a project that has an initial investment of $50,000 and that returns $18,000 per year for the next four years. If the MARR is 12%, is this a good investment?

PW = -50,000 + 18,000 (P/A, 12%, 4)

PW = -50,000 + 18,000 (3.0373)

PW = $4,671.40 → This is a good investment!

Example: What is the value of a 6%, 10-year bond with a par (and redemption) value of $20,000 that pays dividends semi-annually, if the purchaser wishes to earn an 8% return?

V_N *= $20,000 (P/F, 4%, 20) + (0.03)$20,000 (P/A, 4%, 20)*

V_N *= $20,000 (0.4564) + (0.03)$20,000 (13.5903)*

V_N *= $17,282.18*

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5.3. The Present Worth Method

5.3.1. Assumptions of the PW Method

There are several noteworthy assumptions that we make when using PW model.

First, it is assumed that we know the future with certainty (we dont live in a certain world)

Second, it is assumed we can borrow and lend monet at the same interest rate (real World has neither certainity nor perfect)

5.3.2. Bond Value

A bond value is an IOU where you agree to lend the bond issuer money for a specified length of time (say, 10 years). In return, you receive periodic interest payments from issuer plus a promise to return the face value of the bond when it matures.

The balue of a bond, at any time, is the PW of future cash receipts. For a bond,

$$V_N = C(P/F, i\%, N) + rZ(P/A, i\%, N)$$

Z= face, or par, value;
C = redemption or disposal price (usually equals to Z)
r = bond rate (nominal interest) per interest period,
N = number of periods before redemption
i = bond yield rate per period,
V_N = value (price) of the bond N interest periods prior to redemption – this is a PW measure of merit

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5.3. The Present Worth Method

EXAMPLE 5-3 **Stan Moneymaker Wants to Buy a Bond**



Stan Moneymaker has the opportunity to purchase a certain U.S. Treasury bond that matures in eight years and has a face value of $10,000. This means that Stan will receive $10,000 cash when the bond's maturity date is reached. The bond stipulates a fixed nominal interest rate of 8% per year, but interest payments are made to the bondholder every three months; therefore, each payment amounts to 2% of the face value.

Stan would like to earn 10% nominal interest (compounded quarterly) per year on his investment, because interest rates in the economy have risen since the bond was issued. How much should Stan be willing to pay for the bond?

Solution

To establish the value of this bond, in view of the stated conditions, the PW of future cash flows during the next eight years (the study period) must be evaluated. Interest payments are quarterly. Because Stan Moneymaker desires to obtain 10% *nominal interest per year* on the investment, the PW is computed at $i = 10\%/4 = 2.5\%$ per quarter for the remaining $8(4) = 32$ quarters of the bond's life:

$$V_N = \$10{,}000(P/F, 2.5\%, 32) + \$10{,}000(0.02)(P/A, 2.5\%, 32)$$
$$= \$4{,}537.71 + \$4{,}369.84 = \$8{,}907.55.$$

Thus, Stan should pay no more than $8,907.55 when 10% nominal interest per year is desired.

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5.3. The Present Worth Method

EXAMPLE 5-4 **Current Price and Annual Yield of Bond Calculations**

A bond with a face value of \$5,000 pays interest of 8% per year. This bond will be redeemed at par value at the end of its 20-year life, and the first interest payment is due one year from now.

(a) How much should be paid now for this bond in order to receive a yield of 10% per year on the investment?

(b) If this bond is purchased now for \$4,600, what annual yield would the buyer receive?

Solution

(a) By using Equation (5-2), the value of V_N can be determined:

$$V_N = \$5{,}000(P/F, 10\%, 20) + \$5{,}000(0.08)(P/A, 10\%, 20)$$
$$= \$743.00 + \$3{,}405.44 = \$4{,}148.44.$$

(b) Here, we are given $V_N = \$4{,}600$, and we must find the value of $i\%$ in Equation (5-2):

$$\$4{,}600 = \$5{,}000(P/F, i'\%, 20) + \$5{,}000(0.08)(P/A, i'\%, 20).$$

To solve for $i'\%$, we can resort to an iterative trial-and-error procedure (e.g., try 8.5%, 9.0%), to determine that $i'\% = 8.9\%$ per year.

Chapter 5
5.3. The Present Worth Method

5.3.3. The Capitalized-Worth Method

Capitalized worth is a special variation of present worth.

- Capitalized worth is the present worth of all revenues or expenses over an infinite length of time.
- If only expenses are considered this is sometimes referred to as *capitalized cost*.
- The capitalized worth method is especially useful in problems involving endowments and public projects with indefinite lives.

The application of CW concepts.

The CW of a series of end-of-period uniform payments A, with interest at *i%* per period, is *A(P/A, i%, N)*.

As *N* becomes very large (if the *A* are perpetual payments), the (P/A) term approaches *1/i*. So, *CW = A(1/i)*.

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5.3. The Present Worth Method

Example:
Betty has decided to donate some funds to her local community college. Betty would like to fund an endowment that will provide a scholarship of $25,000 each year in perpetuity, and also a special award, "Student of the Decade," each ten years (again, in perpetuity) in the amount of $50,000. How much money does Betty need to donate today, in one lump sum, to fund the endowment? Assume the fund will earn a return of 8% per year.

Solution:
First, convert "Student of the Decade" funds into an equivalent annual amount over the ten years.

$$A = \$50,000(A/F, 8\%, 10) = \$3,451.47$$

We can add this to the annual scholarship and then use capitalized worth to bring it all back to time zero.

$$CW = \frac{\$25,000 + \$3,451.47}{0.08} = \$355,643.43$$

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5.3. The Present Worth Method

EXAMPLE 5-5 **Determining the Capitalized Worth of a Bridge**



A new bridge across the Cumberland River is being planned near a busy highway intersection in the commercial part of a mid-western town. The construction (first) cost of the bridge is $1,900,000 and annual upkeep is estimated to be $25,000. In addition to annual upkeep, major maintenance work is anticipated every eight years at a cost of $350,000 per occurrence. The town government's MARR is 8% per year.

(a) For this problem, what analysis period (N) is, practically speaking, defined as forever?

(b) If the bridge has an expected life of 50 years, what is the capitalized worth (CW) of the bridge over a 100-year study period?

Solution

(a) A practical approximation of "forever" (infinity) is dependent on the interest rate. By examining the $(A/P, i\%, N)$ factor as N increases in the Appendix C tables, we observe that this factor approaches a value of i as N becomes large. For $i = 8\%$ (Table C-11), the $(A/P, 8\%, 100)$ factor is 0.08. So $N = 100$ years is, for practical purposes, "forever" in this example.

(b) The CW is determined as follows:

$$\begin{aligned} CW(8\%) = &-\$1{,}900{,}000 - \$1{,}900{,}000\,(P/F, 8\%, 50) \\ &- [\$350{,}000\,(A/F, 8\%, 8)]/0.08 - \$25{,}000/0.08. \end{aligned}$$

The CW turns out to be −$2,664,220 over a 100-year study period, assuming the bridge is replaced at the end of year 50 for $1,900,000.

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5.4. The Future Worth Method

Future Worth (FW) method is an alternative to the PW method.

- Looking at FW is appropriate since the primary objective is to maximize the future wealth of owners of the firm.
- FW is based on the equivalent worth of all cash inflows and outflows at the end of the study period at an interest rate that is generally the MARR.
- Decisions made using FW and PW will be the same.

FW Decision Rule:

If FW (i=MARR) ≥ 0, the project is economically justified

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5.4. The Future Worth Method

Example:

A $45,000 investment in a new conveyor system is projected to improve throughput and increasing revenue by $14,000 per year for five years. The conveyor will have an estimated market value of $4,000 at the end of five years. Using FW and a MARR of 12%, is this a good investment?

Solution:

FW = -$45,000(F/P, 12%, 5)+$14,000(F/A, 12%, 5)+$4,000

FW = -$45,000(1.7623)+$14,000(6.3528)+$4,000

FW = $13,635.70 → This is a good investment!

Chapter 5
5.4. The Future Worth Method

EXAMPLE 5-6 **The Relationship between FW and PW**

Evaluate the FW of the potential improvement project described in Example 5-1. Show the relationship between FW and PW for this example.

Solution

$$\begin{aligned} FW(20\%) &= -\$25{,}000(F/P, 20\%, 5) \\ &\quad + \$8{,}000(F/A, 20\%, 5) + \$5{,}000 \\ &= \$2{,}324.80. \end{aligned}$$

Again, the project is shown to be a good investment (FW $\geq$ 0). The PW is a multiple of the equivalent FW value:

$$PW(20\%) = \$2{,}324.80(P/F, 20\%, 5) = \$934.29.$$

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5.4. The Future Worth Method

EXAMPLE 5-7 **Sensitivity Analysis Using FW (Example 5-2 Revisited)**



In Example 5-2, the $110,000 retrofitted space-heating system was projected to save $30,000 per year in electrical power and be worth $8,000 at the end of the six-year study period. Use the FW method to determine whether the project is still economically justified if the system has zero market value after six years. The MARR is 15% per year.

Solution

In this example, we need to find the future equivalent of the $110,000 investment and the $30,000 annual savings at an interest rate of 15% per year.

$$\begin{aligned} FW(15\%) &= -\$110{,}000\,(F/P, 15\%, 6) + \$30{,}000\,(F/A, 15\%, 6) \\ &= -\$110{,}000\,(2.3131) + \$30{,}000\,(8.7537) \\ &= \$8{,}170. \end{aligned}$$

The heating system is still a profitable project ($FW \geq 0$) even if it has no market value at the end of the study period.

Chapter 5
5.5. The Annual Worth Method

Annual Worth (AW) is another way to assess projects.

- Annual worth is an equal periodic series of dollar amounts that is *equivalent* to the cash inflows and outflows, at an interest rate that is generally the MARR.

- The AW of a project is annual equivalent revenue or savings (R) minus annual equivalent expenses (E), less its annual capital recovery (*CR*) amount.

$$AW(i\%) = \underline{R} - \underline{E} - CR(i\%)$$

AW Decision Rule:

If AW (i=MARR) ≥ 0, the project is economically justified

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5.5. The Annual Worth Method

Capital recovery reflects the capital cost of the asset.

- *CR* is the annual equivalent cost of the capital invested.

- The *CR* covers the following items.
  - Loss in value of the asset.
  - Interest on invested capital (at the MARR).

- The *CR* distributes the initial cost (*I*) and the salvage value (*S*) across the life of the asset.

$$CR(i\%) = I(A/P, i\%, N) - S(A/F, i\%, N)$$

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5.5. The Annual Worth Method

Example:

A project requires an initial investment of $45,000, has a salvage value of $12,000 after six years, incurs annual expenses of $6,000, and provides an annual revenue of $18,000. Using a MARR of 10%, determine the AW of this project.

$$AW(10\%) = \underline{R} - \underline{E} - CR(10\%)$$

$$CR(10\%) = 45,000(A/P, 10\%, 6) - 12,000(A/F, 10\%, 6)$$

$$CR(10\%) = 8,777$$

$$AW(10\%) = 18,000 - 6,000 - 8,777 = \$3,223$$

Since the *AW* is positive, it's a good investment.

Chapter 5
5.5. The Annual Worth Method

EXAMPLE 5-8 **Using AW to Evaluate the Purchase of New Equipment (Example 5-1 Revisited)**

By using the AW method and Equation (5-4), determine whether the equipment described in Example 5-1 should be recommended.

Solution

The AW Method applied to Example 5-1 yields the following:

$$\begin{aligned} AW(20\%) &= \overbrace{\$8{,}000}^{R-E} - \overbrace{[\$25{,}000(A/P, 20\%, 5) - \$5{,}000(A/F, 20\%, 5)]}^{\text{CR amount [Equation (5-5)]}} \\ &= \$8{,}000 - \$8{,}359.50 + \$671.90 \\ &= \$312.40. \end{aligned}$$

Chapter 5
5.5. The Annual Worth Method

EXAMPLE 5-9 **Equivalent Uniform Annual Cost of a Corporate Jet**

A corporate jet costs $1,350,000 and will incur $200,000 per year in fixed costs (maintenance, licenses, insurance, and hangar rental) and $277 per hour in variable costs (fuel, pilot expense, etc.). The jet will be operated for 1,200 hours per year for five years and then sold for $650,000. The MARR is 15% per year.

(a) Determine the capital recovery cost of the jet.

(b) What is the EUAC of the jet?

Solution

(a) $CR = \$1{,}350{,}000\,(A/P, 15\%, 5) - \$650{,}000\,(A/F, 15\%, 5) = \$306{,}310.$

(b) The total annual expense for the jet is the sum of the fixed costs and the variable costs.

$$\underline{E} = \$200{,}000 + (1{,}200 \text{ hours})(\$277/\text{hour}) = \$532{,}400$$

$$\text{EUAC}(15\%) = \$532{,}400 + \$306{,}310 = \$838{,}710$$

Chapter 5
5.5. The Annual Worth Method

EXAMPLE 5-10 **Determination of Annual Savings by Using the AW Method (Example 5-2 Revisited)**



Consider the retrofitted space-heating system described in Example 5-2. Given the investment of $110,000 and market value of $8,000 at the end of the six-year study period, what is the minimum annual electrical power savings (in kWh) required to make this project economically acceptable? The MARR = 15% per year and electricity costs $0.10 per kWh.

Solution

To make this project acceptable, the annual power savings must be at least as great as the annual CR amount. Using Equation (5-5),

$$CR = \$110{,}000(A/P, 15\%, 6) - \$8{,}000\,(A/F, 15\%, 6) = \$28{,}148.40.$$

This value ($28,148.40) is the minimum annual dollar savings needed to justify the space-heating system. This equates to

$$\frac{\$28{,}148.40}{\$0.10/\text{kWh}} = 281{,}480 \text{ kWh per year.}$$

If the space-heating system can save 281,480 kWh per year, it is economically justified (exactly 15% is earned on the beginning-of-year unrecovered investment). Any savings greater than 281,480 kWh per year (such as the original estimate of 300,000 kWh per year) will serve to make this project even more attractive.

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5.5. The Annual Worth Method

EXAMPLE 5-11 **Avoid Getting Fleeced on an Auto Lease**

Automobile leases are built around three factors: negotiated sales price, residual value, and interest rate. The residual value is what the dealership expects the car's value will be when the vehicle is returned at the end of the lease period. The monthly cost of the lease is the capital recovery amount determined by using these three factors.

(a) Determine the monthly lease payment for a car that has an agreed-upon sales price of \$34,995, an APR of 9% compounded monthly, and an estimated residual value of \$20,000 at the end of a 36-month lease.* An up-front payment of \$3,000 is due when the lease agreement (contract) is signed.

(b) If the estimated residual value is raised to \$25,000 by the dealership to get your business, how much will the monthly payment be?

Solution

(a) The effective sales price is \$31,995 (\$34,995 less the \$3,000 due at signing). The monthly interest rate is 9%/12 = 0.75% per month. So the capital recovery amount is:

$$\begin{aligned} CR &= \$31{,}995(A/P, 0.75\%, 36) - \$20{,}000(A/F, 0.75\%, 36) \\ &= \$1{,}017.44 - \$486 \\ &= \$531.44 \text{ per month.} \end{aligned}$$

* There are many excellent Web sites offering free calculators for car loans, home mortgages, life insurance, and credit cards. In this regard, check out www.choosetosave.org.

Chapter 5
5.6. The Internal Rate of Return Method

- The internal rate of return (IRR) method is the most widely used rate of return method for performing engineering economic analysis.
- It is also called the *investor's method*, the *discounted cash flow* method, and the *profitability index*.
- If the IRR for a project is greater than the MARR, then the project is *acceptable*.

How the IRR works

- The IRR is the interest rate that equates the equivalent worth of an alternative's cash *inflows* (revenue, *R*) to the equivalent worth of cash *outflows* (expenses, *E*).
- The IRR is sometimes referred to as the *breakeven interest rate*.

The IRR is the interest *i'%* at which

$$\sum_{k=0}^{N} R_k(P/F, i'\%, k) = \sum_{k=0}^{N} E_k(P/F, i'\%, k)$$

IRR Decision Rule:

If IRR (i=MARR) ≥ 0, the project is economically justified

Chapter 5
5.6. The Internal Rate of Return Method

Solving for the IRR is a bit more complicated than PW, FW, or AW

- The method of solving for the *i'%* that equates revenues and expenses normally involves trial-and-error calculations, or solving numerically using mathematical software.
- The use of spreadsheet software can greatly assist in solving for the IRR. Excel uses the *IRR(range, guess)* or *RATE(nper, pmt, pv)* functions.

Challenges in applying the IRR method.

- It is computationally difficult without proper tools.
- In rare instances multiple rates of return can be found. (See Appendix 5-A.)
- The IRR method must be carefully applied and interpreted when comparing two more mutually exclusive alternatives (e.g., do not directly compare internal rates of return).

Chapter 5
5.6. The Internal Rate of Return Method

EXAMPLE 5-14 **Do You Know What Your Effective Interest Rate Is?**



In 1915, Albert Epstein allegedly borrowed \$7,000 from a large New York bank on the condition that he would repay 7% of the loan every three months, until a total of 50 payments had been made. At the time of the 50th payment, the \$7,000 loan would be completely repaid. Albert computed his annual interest rate to be $[0.07(\$7{,}000) \times 4]/\$7{,}000 = 0.28$ (28%).

(a) What true *effective* annual interest rate did Albert pay?

(b) What, if anything, was wrong with his calculation?

Solution

(a) The true interest rate per quarter is found by equating the equivalent value of the amount borrowed to the equivalent value of the amounts repaid. Equating the AW amounts per quarter, we find

$$\$7{,}000(A/P, i'\%/\text{quarter}, 50 \text{ quarters}) = 0.07(\$7{,}000) \text{ per quarter},$$

$$(A/P, i'\%, 50) = 0.07.$$

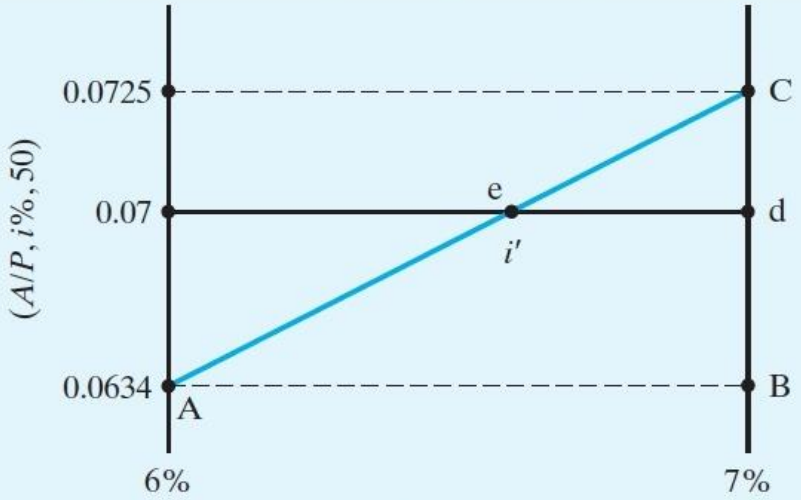
Linearly interpolating to find $i'\%$/quarter by using similar triangles is the next step:

$$(A/P, 6\%, 50) = 0.0634,$$

$$(A/P, 7\%, 50) = 0.0725.$$

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5.6. The Internal Rate of Return Method

EXAMPLE 5-14 **Do You Know What Your Effective Interest Rate Is?**



$$\frac{\text{line de}}{\text{line BA}} = \frac{\text{line Cd}}{\text{line CB}}$$

$$\frac{7\% - i'\%}{7\% - 6\%} = \frac{0.0725 - 0.07}{0.0725 - 0.0634},$$

$$i'\% = 7\% - 1\% \left(\frac{0.0025}{0.0091}\right),$$

or $i'\% \simeq 6.73\%$ per quarter.

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5.6. The Internal Rate of Return Method

EXAMPLE 5-14 **Do You Know What Your Effective Interest Rate Is?**

Now we can compute the effective $i'\%$ per year that Albert was paying:

$$i'\% = [(1.0673)^4 - 1]100\%$$

$$\simeq 30\% \text{ per year.}$$

(b) Even though Albert's answer of 28% is close to the true value of 30%, his calculation is insensitive to how long his payments were made. For instance, he would get 28% for an answer when 20, 50, or 70 quarterly payments of \$490 were made! For 20 quarterly payments, the true effective interest rate is 14.5% per year, and for 70 quarterly payments, it is 31% per year. As more payments are made, the true annual effective interest rate being charged by the bank will increase, but Albert's method would not reveal by how much.

Another Solution

When the initial loan amount, the payment amount, and the number of payments are known, Excel has a useful financial function, RATE (*nper*, *pmt*, *pv*), that will return the interest rate per period. For this example,

$$\text{RATE}(50, -490, 7000) = 6.73\%.$$

This is the same quarterly interest rate we obtained via linear interpolation in Part (a).

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5.6. The Internal Rate of Return Method

EXAMPLE 5-15 **Be Careful with "Fly-by-Night" Financing!**

The Fly-by-Night finance company advertises a "bargain 6% plan" for financing the purchase of automobiles. To the amount of the loan being financed, 6% is added for each year money is owed. This total is then divided by the number of months over which the payments are to be made, and the result is the amount of the monthly payments. For example, a woman purchases a $10,000 automobile under this plan and makes an initial cash payment of $2,500. She wishes to pay the $7,500 balance in 24 monthly payments:

Purchase price	=	$10,000
−Initial payment	=	2,500
= Balance due, (P_0)	=	7,500
+6% finance charge = 0.06 × 2 years × $7,500	=	900
= Total to be paid	=	8,400
∴ Monthly payments (A) = $8,400/24	=	$350

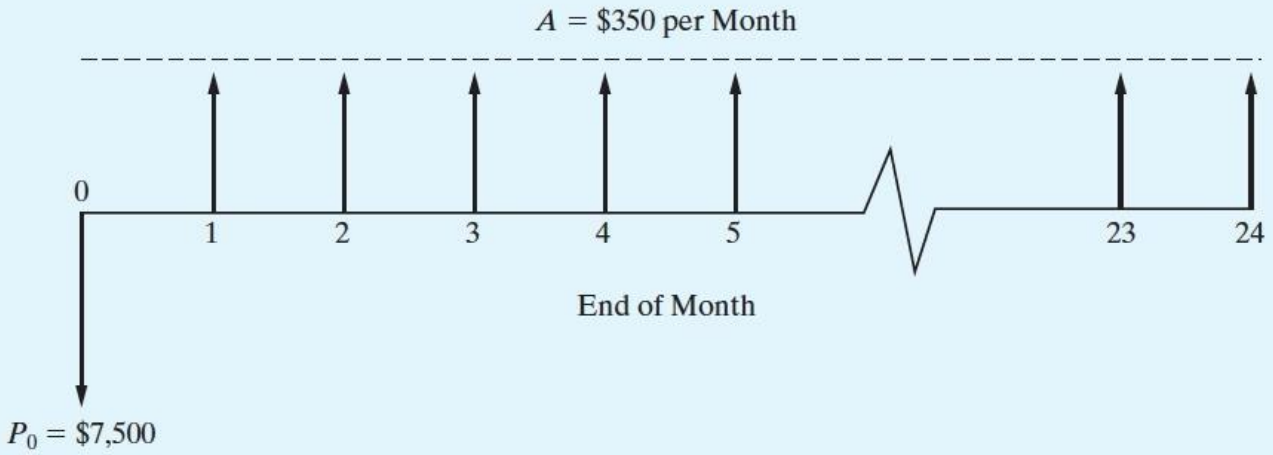
What effective annual rate of interest does she actually pay?

Chapter 5
5.6. The Internal Rate of Return Method

EXAMPLE 5-15 **Be Careful with "Fly-by-Night" Financing!**

Solution

Because there are to be 24 payments of \$350 each, made at the end of each month, these constitute an annuity (A) at some unknown rate of interest, $i'\%$, that should be computed only upon the unpaid balance instead of on the entire \$7,500 borrowed. A cash-flow diagram of this situation is shown below.



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5.6. The Internal Rate of Return Method

EXAMPLE 5-15 **Be Careful with "Fly-by-Night" Financing!**

In this example, the amount owed on the automobile (i.e., the initial unpaid balance) is \$7,500, so the following equivalence expression is employed to compute the unknown monthly interest rate:

$$P_0 = A(P/A, i'\%, N),$$

$$\$7{,}500 = \$350/\text{month}(P/A, i'\%, 24 \text{ months}),$$

$$(P/A, i'\%, 24) = \frac{\$7{,}500}{\$350} = 21.43.$$

Examining the interest tables for P/A factors at $N = 24$ that come closest to 21.43, we find that $(P/A, 3/4\%, 24) = 21.8891$ and $(P/A, 1\%, 24) = 21.2434$.

Using the linear interpolation procedure, the IRR is computed as 0.93% per month*, since payments are monthly. The nominal rate paid on the borrowed money is $0.93\%(12) = 11.16\%$ compounded monthly. This corresponds to an effective annual interest rate of $[(1 + 0.0093)^{12} - 1] \times 100\% \cong 12\%$. What appeared at first to be a real bargain turns out to involve effective annual interest at twice the stated rate. The reason is that, on the average, only \$3,750 is borrowed over the two-year period, but interest on \$7,500 over 24 months was charged by the finance company.

* Using Excel, RATE(24, −350, 7500) = 0.93%.

Chapter 5
5.6. The Internal Rate of Return Method

EXAMPLE 5-16 **Effective Interest Rate for Purchase of New Aircraft Equipment**

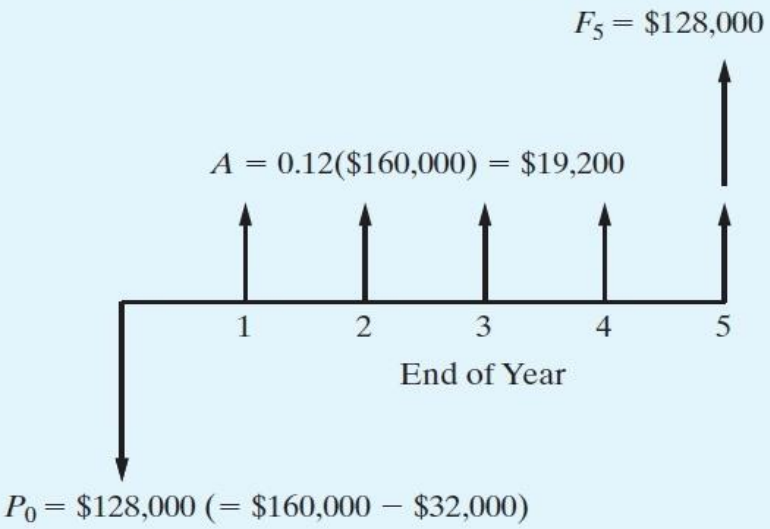
A small airline executive charter company needs to borrow $160,000 to purchase a prototype synthetic vision system for one of its business jets. The SVS is intended to improve the pilots' situational awareness when visibility is impaired. The local (and only) banker makes this statement: "We can loan you $160,000 at a very favorable rate of 12% per year for a five-year loan. However, to secure this loan, you must agree to establish a checking account (with no interest) in which the *minimum* average balance is $32,000. In addition, your interest payments are due at the end of each year, and the principal will be repaid in a lump-sum amount at the end of year five." What is the true effective annual interest rate being charged?

Solution

The cash-flow diagram from the banker's viewpoint appears below. When solving for an unknown interest rate, it is good practice to draw a *cash-flow diagram* prior to writing an equivalence relationship. Notice that $P_0 = \$160{,}000 - \$32{,}000 = \$128{,}000$. Because the bank is requiring the company to open an account worth $32,000, the bank is only $128,000 out of pocket. This same principal applies to F_5 in that the company only needs to repay $128,000 since the $32,000 on deposit can be used to repay the original principal.

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5.6. The Internal Rate of Return Method

EXAMPLE 5-16 **Effective Interest Rate for Purchase of New Aircraft Equipment**



The interest rate (IRR) that establishes equivalence between positive and negative cash flows can now easily be computed:*

$$P_0 = F_5(P/F, i'\%, 5) + A(P/A, i'\%, 5),$$

$$\$128{,}000 = \$128{,}000(P/F, i'\%, 5) + \$19{,}200(P/A, i'\%, 5).$$

If we try $i' = 15\%$, we discover that \$128,000 = \$128,000. Therefore, the true effective interest rate is 15% per year.

* Several Web sites provide excellent tutorials for equivalent worth (e.g., PW) and rate-of-return calculations. For example, see www.investopedia.com and www.datadynamica.com/IRR.asp.

Chapter 5
5.7. The External Rate of Return Method

Reinvesting revenue—the External Rate of Return (ERR)

- The IRR assumes revenues generated are reinvested at the IRR—which may not be an accurate situation.
- The ERR takes into account the interest rate, ε, external to a project at which net cash flows generated (or required) by a project over its life can be reinvested (or borrowed). This is usually the MARR.
- If the ERR happens to equal the project's IRR, then using the ERR and IRR produce identical results.

The ERR procedure

- Discount all the net cash *outflows* to time *0* at ε% per compounding period.
- Compound all the net cash *inflows* to period *N* at at ε%.
- Solve for the ERR, the interest rate that establishes equivalence between the two quantities.

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5.7. The External Rate of Return Method

ERR is the *i'%* at which

$$\sum_{k=0}^{N} E_k(P/F, \varepsilon\%, k)(F/P, i'\%, N) = \sum_{k=0}^{N} R_k(F/P, \varepsilon\%, N-k)$$

where

R_k = excess of receipts over expenses in period *k*,
E_k = excess of expenses over receipts in period *k*,
N = project life or number of periods, and
ε = external reinvestment rate per period.

Chapter 5
5.7. The External Rate of Return Method

Applying the ERR method

For the cash flows given below, find the ERR when the external reinvestment rate is ε = 12% (equal to the MARR).

Year	0	1	2	3	4
Cash Flow	-$15,000	-$7,000	$10,000	$10,000	$10,000

Expenses $15,000 + 7,000(P/F, 12\%, 1) = 21,250$

Revenue $10,000(F/A, 12\%, 3) = 33,744$

Solving, we find

$$21,250(F/P, i'\%, 4) = 33,744$$

$$i' = 16.67\% > 12\%$$

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5.7. The External Rate of Return Method

EXAMPLE 5-17 **Calculation of the ERR**

Referring to Example 5-13, suppose that $\epsilon = \text{MARR} = 20\%$ per year. What is the project's ERR, and is the project acceptable?

Solution

By utilizing Equation (5-8), we have the following relationship to solve for i':

$$\$25{,}000(F/P, i'\%, 5) = \$8{,}000(F/A, 20\%, 5) + \$5{,}000,$$

$$(F/P, i'\%, 5) = \frac{\$64{,}532.80}{\$25{,}000} = 2.5813 = (1 + i')^5,$$

$$i' = 20.88\%.$$

Because $i' >$ MARR, the project is justified, but just barely.

Chapter 5
5.7. The External Rate of Return Method

EXAMPLE 5-17 **Calculation of the ERR**

Referring to Example 5-13, suppose that $\epsilon = \text{MARR} = 20\%$ per year. What is the project's ERR, and is the project acceptable?

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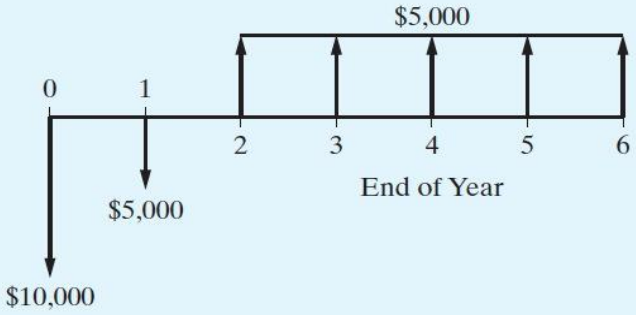
Because $i' >$ MARR, the project is justified, but just barely.

Chapter 5
5.7. The External Rate of Return Method

EXAMPLE 5-18 **Determining the Acceptability of a Project, Using ERR**

VIDEO SOLUTIONS

When $\epsilon = 15\%$ and MARR $= 20\%$ per year, determine whether the project (whose net cash-flow diagram appears next) is acceptable. Notice in this example that the use of an $\epsilon\%$ different from the MARR is illustrated. This might occur if, for some reason, part or all of the funds related to a project are "handled" outside the firm's normal capital structure.



Solution

$$E_0 = \$10{,}000 \ (k = 0),$$
$$E_1 = \$5{,}000 \ (k = 1),$$
$$R_k = \$5{,}000 \text{ for } k = 2, 3, \ldots, 6,$$

$$[\$10{,}000 + \$5{,}000(P/F, 15\%, 1)](F/P, i'\%, 6) = \$5{,}000(F/A, 15\%, 5);$$
$$i'\% = 15.3\%.$$

The $i'\%$ is less than the MARR $= 20\%$; therefore, this project would be unacceptable according to the ERR method.

Chapter 5
5.8. The Payback (Payout) Period Method

The payback period method is simple, but possibly misleading.

- The simple payback period is the number of years required for cash *inflows* to just equal cash *outflows*.
- It is a measure of *liquidity* rather than a measure of profitability.

Payback is simple to calculate.

The payback period is the *smallest* value of θ ($\theta \leq N$) for which the relationship below is satisfied.

$$\sum_{k=1}^{\theta}(R_k - E_k) - I \geq 0$$

For *discounted* payback future cash flows are discounted back to the present, so the relationship to satisfy becomes

$$\sum_{k=1}^{\theta'}(R_k - E_k)(P/F, i\%, k) - I \geq 0$$

Chapter 5
5.8. The Payback (Payout) Period Method

Problems with the payback period method.

- It doesn't reflect any cash flows occurring after θ, or θ'.
- It doesn't indicate anything about project desirability except the speed with which the initial investment is recovered.
- Recommendation: use the payback period only as supplemental information in conjunction with one or more of the other methods in this chapter.

Chapter 5
5.8. The Payback (Payout) Period Method

Finding the simple and discounted payback period for a set of cash flows.

The cumulative cash flows in the table were calculated using the formulas for simple and discounted payback.

From the calculations $\theta = 4$ years and $\theta' = 5$ years.

End of Year	Net Cash Flow	Cumulative PW at *0%*	Cumulative PW at *6%*
0	-$42,000	-$42,000	-$42,000
1	$12,000	-$30,000	-$30,679
2	$11,000	-$19,000	-$20,889
3	$10,000	-$9,000	-$12,493
4	$10,000	$1,000	-$4,572
5	$9,000		$2,153

Chapter 5
5.8. The Payback (Payout) Period Method

EXAMPLE 5-19 **Determining the Simple Payback Period**



A public school is being renovated for $13.5 million. The building has geothermal heating and cooling, high-efficiency windows, and a solar array that permits the school to sell electricity back to the local electric utility. The annual value of these benefits is estimated to be $2.7 million. In addition, the residual value of the school at the end of its 40-year life is negligible. What is the simple payback period and internal rate of return for the renovated school?

Solution

The simple payback period is

$$\frac{\$13.5 \text{ million}}{\$2.7 \text{ million/year}} = 5 \text{ years.}$$

This is fairly good for a publically sponsored project. The IRR can be computed using the equation

$$0 = -\$13.5 \text{ million} + \$2.7 \text{ million } (P/A, i'\%, 40),$$

yielding $i'\%$ (the IRR) $= 20\%$ per year. The IRR indicates the project is profitable for a MARR of 20% per year or less.